Saintgits College of Engineering (Autonomous)

Department of Electronics and Communication Engineering

# Transmission Line Signal Integrity Using TDR

Capstone Project — Problem B

Computational Project Report

**Course:** 24BSE2113-D — Partial Differential Equations, Transforms and Optimization Techniques

**Group 6**

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**Abstract**

This project develops a computational workflow for locating a localized transmission-line fault from time-domain reflectometry (TDR). A synthetic 1 m line is described by distributed resistance, inductance, conductance and capacitance. A three-node fault near the midpoint increases resistance to four times nominal and reduces capacitance to 60% of nominal. The coupled Telegrapher equations provide the forward model, a staggered finite-difference time-domain (FDTD) scheme generates nominal and faulty traces, and a sensitivity matrix supports a nonnegative L1 linear-programming estimate. The selected time step satisfies the CFL-type stability bound and the HiGHS solver converges. However, the strongest reconstructed anomaly is at 0.4286 m while the true centre is 0.4898 m, an error of three grid cells. Thus the pipeline runs end to end but does not meet the required two-cell localization tolerance. The study is a synthetic proof of concept, not validation on measured cable data.

# 1 Introduction

Transmission lines carry fast electrical signals in communication systems, circuit interconnects and instrumentation. A local impedance change disturbs the travelling wave and produces a reflected voltage. Time-domain reflectometry applies a known transition and examines the response to estimate the location of a discontinuity. This project asks whether a localized change in distributed line parameters can be identified from a simulated input trace.

The model is a 1 m line with nominal parameters $R_0 = 0.1\,\Omega/\mathrm{m}$, $L_0 = 250\,\mathrm{nH/m}$, $C_0 = 100\,\mathrm{pF/m}$ and $G_0 = 0\,\mathrm{S/m}$. Three adjacent nodes near the midpoint are modified, with resistance set to $4R_0$ and capacitance to $0.6C_0$. The workflow connects field-based line equations to numerical simulation and then inverse optimization. Accuracy is judged against the known synthetic fault rather than inferred from solver success alone.

# 2 Related Work

Paul's treatment of transmission lines establishes the voltage-current equations and their time- and frequency-domain interpretation [1]. These relations provide the physical basis of the present model, although the standard analytical results are most straightforward for uniform lines and do not themselves estimate unknown spatial parameters. Taflove and Hagness describe finite-difference time-domain methods and their stability requirements [2]; this project adapts the staggered-grid principle to a one-dimensional distributed RLCG line. Boyd and Vandenberghe explain convex optimization, including linear formulations for constrained fitting [3]. SciPy exposes linear programming through `linprog`, which can use the HiGHS solver [4, 5].

The methods are established; the small contribution here is an integrated test of a spatially localized, simultaneous resistance and capacitance fault using a sensitivity-based inverse model. The remaining gap in this implementation is not the absence of a

solver, but whether a first-order linearization can localize a relatively large fault within the required tolerance. The experiment explicitly reports this error.

# 3 Methodology

Let $V(x,t)$ and $I(x,t)$ denote voltage and current at position $x$ and time $t$. The per-unit-length parameters $R, L, G, C$ represent series resistance, series inductance, shunt conductance and shunt capacitance. The Telegrapher equations are

$$\frac{\partial V}{\partial x} = -RI - L\frac{\partial I}{\partial t}, \tag{1}$$

$$\frac{\partial I}{\partial x} = -GV - C\frac{\partial V}{\partial t}. \tag{2}$$

The first equation accounts for resistive voltage drop and inductive response; the second describes leakage and capacitive charging. For locally constant parameters they combine to

$$\frac{\partial^2 V}{\partial x^2} = LC\frac{\partial^2 V}{\partial t^2} + (RC + LG)\frac{\partial V}{\partial t} + RGV.$$

The coupled first-order equations are retained numerically because the line parameters vary with position.

With zero initial conditions, the Laplace transform replaces time differentiation by multiplication by complex frequency $s$:

$$\frac{dV}{dx} = -(R + Ls)I, \qquad \frac{dI}{dx} = -(G + Cs)V.$$

Thus the propagation constant and characteristic impedance are

$$\gamma(s) = \sqrt{(R + Ls)(G + Cs)}, \qquad Z_0(s) = \sqrt{\frac{R + Ls}{G + Cs}}.$$

Here $\gamma$ describes propagation and attenuation per unit length, while $Z_0$ is the travelling-wave voltage/current ratio. For the nominal lossless check, $Z_0 = \sqrt{L_0/C_0} = 50\,\Omega$ and wave speed $v = 1/\sqrt{L_0 C_0} = 2.0 \times 10^8\,\mathrm{m/s}$.

The 1 m line is discretized into $N_x = 50$ voltage nodes, so $\Delta x = \ell/(N_x - 1) = 0.02041\,\mathrm{m}$. Voltage is stored at nodes and current at half-cell edges. The current update is

$$I_{i+1/2}^{n+1/2} = a_i I_{i+1/2}^{n-1/2} - b_i (V_{i+1}^n - V_i^n),$$

where $i$ is spatial index, $n$ is time index, and

$$a_i = \frac{1 - R_i \Delta t/(2L_i)}{1 + R_i \Delta t/(2L_i)}, \quad b_i = \frac{\Delta t/(L_i \Delta x)}{1 + R_i \Delta t/(2L_i)}.$$

For $G = 0$, the interior voltage is updated by

$$V_i^{n+1} = V_i^n - \frac{\Delta t}{C_i \Delta x}\left(I_{i+1/2}^{n+1/2} - I_{i-1/2}^{n+1/2}\right).$$

The chosen step is $\Delta t = 71.136$ ps, below the CFL-type limit $\Delta t_{\mathrm{CFL}} = \Delta x / v_{\max} = 79.040$ ps. A smooth 1 V step excites the line, and the far end is approximately matched to $50\,\Omega$.

The inverse model defines nonnegative resistance increase and capacitance reduction as

$$p_R = \frac{R - R_0}{R_0} \geq 0, \quad p_C = \frac{C_0 - C}{C_0} \geq 0, \quad p = \begin{bmatrix} p_R \\ p_C \end{bmatrix}.$$

The parameterization makes both prescribed fault changes nonnegative. Each column of sensitivity matrix $A$ is generated by changing one node by 10%, rerunning the forward model and recording the change in sampled response. If $b$ is the faulty-minus-nominal trace, then $b \approx Ap$. There are 98 time samples and 100 candidate parameters, so $A \in \mathbb{R}^{98\times 100}$ and $b \in \mathbb{R}^{98}$.

Absolute residuals are linearized with nonnegative auxiliary variables $\epsilon_k$. The optimization is

$$\min_{p,\epsilon} \sum_{k=1}^{m} \epsilon_k + \lambda \sum_j p_j, \qquad -\epsilon \leq Ap - b \leq \epsilon, \quad p \geq 0, \quad \epsilon \geq 0.$$

Here $m = 98$ is the number of observations, $k$ indexes time samples, $j$ indexes candidate parameters, and $\lambda = 2 \times 10^{-4}$ is a small sparsity penalty. Column scaling balances the magnitudes of resistance and capacitance sensitivities. This converts the fit into a linear program.

# 4 Experimental Setup

The experiment was implemented in Python in a Jupyter notebook. NumPy supports numerical arrays and updates, Matplotlib produces the figures, and SciPy's `scipy.optimize.linprog` with `method='highs'` solves the LP. A fixed random seed makes the added Gaussian noise repeatable. The line uses 50 nodes, $\Delta x = 0.02041$ m and 169 time steps. The fault occupies nodes 23–25, at 0.4694, 0.4898 and 0.5102 m. Sensitivity samples are taken between 1 and 8 ns. The model uses an approximate 50 Ω termination and synthetic data only; no physical cable or oscilloscope measurements were collected.

# 5 Results and Discussion

The nominal and faulty simulated input traces differ after the launched transition. Subtracting the nominal trace highlights the fault signature used in the inverse stage. Figure 1 shows these synthetic traces. The difference is generated by the model, not measured

hardware data.

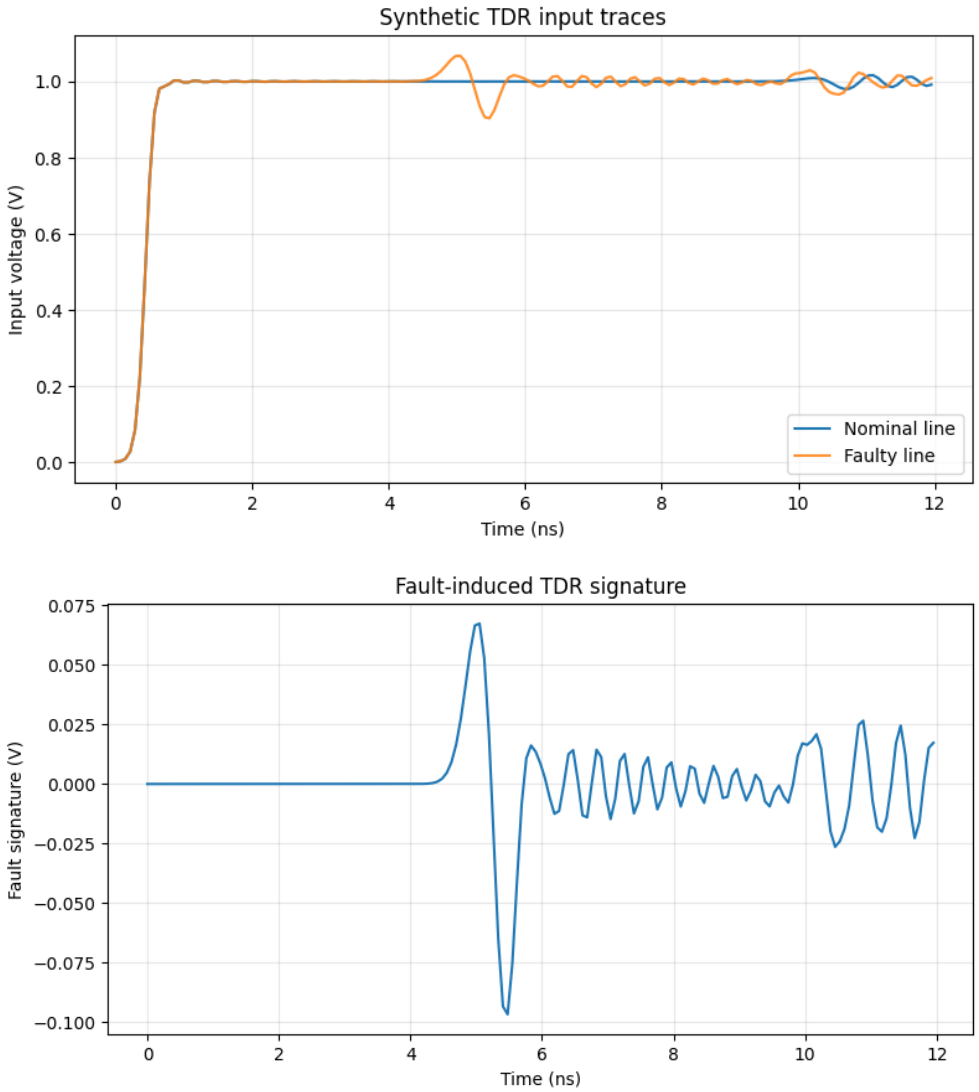



Figure 1: Nominal and faulty synthetic TDR traces and their difference.

The HiGHS solver terminates successfully and reconstructed resistance and capacitance remain nonnegative. Nevertheless, the maximum combined anomaly score $p_R + p_C$ occurs at node 21, or 0.4286 m, while the true fault centre is node 24, or 0.4898 m. The error is three grid cells, which exceeds the two-cell tolerance. Solver convergence therefore demonstrates that the optimization problem was solved, but not that the inferred location is accurate.

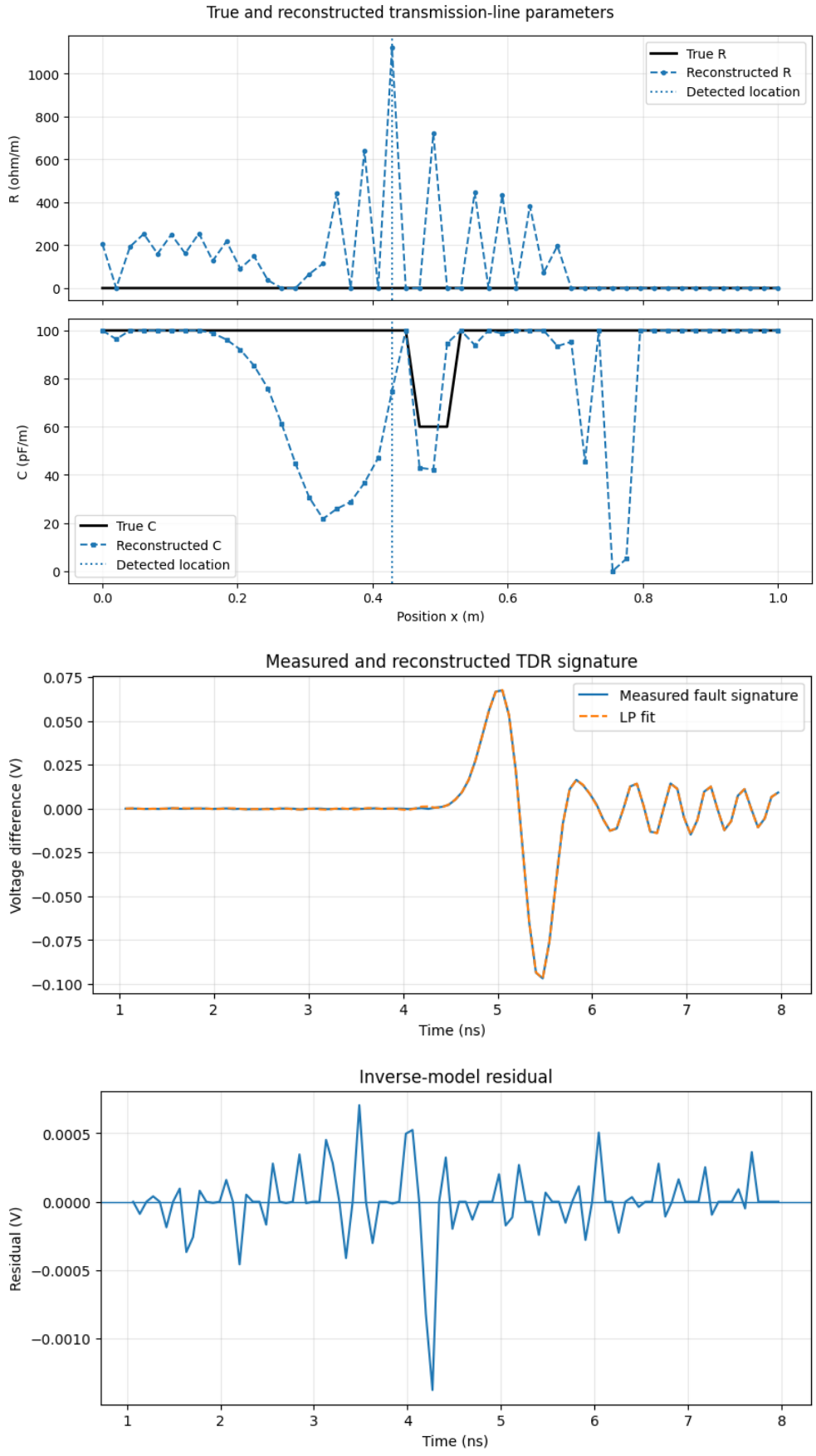



Figure 2: True and reconstructed parameter profiles, the LP fit and the inverse residual. The shifted anomaly and residual indicate mismatch between the linearized model and the imposed fault.

| Verification metric | Result |
|---|---|
| CFL condition | Pass: 71.136 < 79.040 ps |
| LP solver | Successful / optimal |
| Reconstructed $R(x)$ | Nonnegative |
| Reconstructed $C(x)$ | Nonnegative |
| True fault centre | 0.4898 m (node 24) |
| Detected strongest anomaly | 0.4286 m (node 21) |
| Localization error | 3 cells; 2-cell target not met |

Table 1: Numerical verification summary.

The table separates computational feasibility from inference quality. The time-step check and successful LP solve confirm that the computation ran, while the three-cell location error shows that the inverse method requires improvement.

# 6 Limitations

The approximation $b \approx Ap$ is first-order, but the fault is large: resistance is multiplied by four and capacitance is reduced to 60% of nominal. Resistance and capacitance can also produce partially correlated TDR signatures. The source, termination and noise are idealized, and the spatial grid limits resolution. Since the data are synthetic, performance on real cables remains untested.

# 7 Conclusion

The project connects transmission-line modelling, Laplace-domain interpretation, FDTD simulation, sensitivity analysis and nonnegative L1 optimization in a single workflow. The time step satisfies the stability bound, the LP converges, and the reconstructed parameters remain nonnegative. However, the detected anomaly is three grid cells from the true centre, so the two-cell localization target is not achieved. Future work should test iterative re-linearization or nonlinear optimization, tune the sampling window and regularization on validation cases, and evaluate the method using measured TDR data.

# 8 References

# References

[1] C. R. Paul, *Analysis of Multiconductor Transmission Lines*, 2nd ed. Hoboken, NJ: Wiley-IEEE Press, 2008.

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[4] P. Virtanen et al., "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020, doi: 10.1038/s41592-019-0686-2.

[5] SciPy Developers, "scipy.optimize.linprog — Linear programming," SciPy documentation. Available: https://docs.scipy.org/doc/scipy/reference/optimize.linprog.html. Accessed Oct. 12, 2026.

[6] J. C. Maxwell, *A Treatise on Electricity and Magnetism*, vol. 2. Oxford: Clarendon Press, 1873.